UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 010 · Problems on maxima and minima for a function of two variables · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S010·8	Deck 010, slide 8 (rendered .pptx)
T010a 00:07:38	AV-summary transcript "Extremas of functions continued" (older lecture), segment start
T010b 00:03:12	AV-summary transcript "Problems on Maxima and Minima for a function of two variables" (newer lecture)
CW U1 P29	Unit 1 class-work problem 29
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c
ESA Jan-24 s2 1a	Past paper: Jan 2024 set 2, question 1.a

Contents

1. 1 · The method, and the ESA cubic functions
2. 2 · Functions with many stationary points
3. 3 · Applied problems: eliminate one variable first
4. Past-paper questions

1 The method, and the ESA cubic functions

Definitions

Stationary (critical) points. The solutions of $f_x = 0$, $f_y = 0$, solved simultaneously; the points at which the function may have an extreme value. S010·7 · T010a 00:04:13 · T010b 00:01:23

Maximum / minimum value. The value $f(a, b)$ at a point of maximum or minimum, obtained by substituting the point into f. S010·8 · T010b 00:04:02

Statements in the class material

> *"If $rt - s^2 > 0$ and $r < 0$ at (a, b), $f(a, b)$ is a maximum value; if $rt - s^2 > 0$ and $r > 0$, a minimum value; if $rt - s^2 < 0$, $f(a, b)$ is not an extreme value, i.e. (a, b) is a saddle point; if $rt - s^2 = 0$ the case is doubtful and needs further investigation (class 009, restated)."* S010·8 · T010b 00:05:00

Key points

1. Procedure (lecture): find $p = f_x$, $q = f_y$; equate both to zero; find the critical points; compute r, s, t ; evaluate $rt - s^2$ at each critical point; conclude. ESA ★★★ T010b 00:05:00 · T010b 00:05:16
2. Split $f_y = 0$ into cases and substitute each into $f_x = 0$: for the ESA cubic, $y = 0$ gives $x = 4, 6$ and $x = 5$ gives $y = \pm 1$. ESA ★★★ S010·7 · S010·8 · T010b 00:01:40
3. Count the stationary points before classifying; a cubic in x and y typically gives four, a quartic up to nine. ESA ★★ ADDED FOR CLARITY S010·8 · S010·11
4. A saddle point has no value to report; give the maximum and minimum values only. ESA ★★ S010·8
5. When $rt - s^2 > 0$, r and t have the same sign, so either decides between maximum and minimum. ESA ★ ADDED FOR CLARITY S010·8

Why?

Why does $f_y = 6y(x - 5) = 0$ give two separate cases?
A product is zero when either factor is zero: $y = 0$ or $x = 5$. Each case is substituted into $f_x = 0$ to find its own points; missing a case loses stationary points.

ADDED FOR CLARITY S010·7 · T010b 00:01:40

Exam traps

Exam trap. Slide 010·8 says the maximum value is at $(6, 0)$; the maximum 112 is at $(4, 0)$, the minimum 108 at $(6, 0)$. S010·8 · T010b 00:04:42

Exam trap. $rt - s^2 = 36(x - 5)^2 - 36y^2$: the 36 multiplies y^2 too (slide 010·8 drops it).

ADDED FOR CLARITY S010·8

2 Functions with many stationary points

Key points

1. Separable functions such as $x^4 + y^4 - x^2 - y^2 + 1$ give f_x in x only and f_y in y only: combine every root of one with every root of the other. ESA ★★ S010·11 · T010b 00:08:29
2. For $\sin x + \sin y + \sin(x + y)$, subtract the two conditions to get $\cos x = \cos y$, i.e. $x = y$ in the working range; then $\cos 2x = -\cos x = \cos(\pi - x)$ gives $x = \frac{\pi}{3}$. ESA ★★ S010·13 · T010b 00:10:23

Exam traps

Exam trap. When $rt - s^2 < 0$ the sign of r is irrelevant; at $(\pm\frac{1}{\sqrt{2}}, 0)$, $r = 4$ (not -2 as slide 010·12 says), and the point is still a saddle. ADDED FOR CLARITY S010·12 · T010b 00:09:26

Exam trap. At $(\frac{\pi}{3}, \frac{\pi}{3})$, $s = -\frac{\sqrt{3}}{2}$, so $rt - s^2 = 3 - \frac{3}{4} = \frac{9}{4}$ (slide 010·14 prints $\frac{3}{4}$).
ADDED FOR CLARITY S010·14

3 Applied problems: eliminate one variable first

Definitions

Elimination. Using the given condition (constraint) to express one variable in terms of the others, so that the quantity becomes a function of two independent variables. T010a 00:00:40 · T010a 00:08:27

Key points

1. We cannot yet find extrema of a function of three variables, so use the constraint to eliminate one (here z^2 or z). ESA ★★ T010a 00:00:40 · T010a 00:08:27
2. A product such as $kxyz^2$ is largest when x and y have the same sign and $k > 0$; points with x, y of opposite signs give the lowest temperature. ESA ★★ S010·10 · T010b 00:05:40
3. Reject stationary points that make no physical sense (an edge $y = 0$ in the box problem makes $x \to \infty$). ESA ★ T010a 00:13:07 · T010a 00:13:48
4. Elimination is not always easy; class 011's Lagrange multipliers avoid it (T010a 00:16:49). ESA ★ T010a 00:16:49

Why?

Why is a cube the wrong answer for an open box?
Without a top the base is used once and each side face twice in the area, so the box should be lower and wider: $x = y = \sqrt[3]{20}$, $z = \frac{1}{2}\sqrt[3]{20}$. The cube $x = y = z = \sqrt[3]{V}$ minimises the area of a closed box $2(xy + yz + zx)$.
ADDED FOR CLARITY S010·6 · CW U1 P34

Exam traps

Exam trap. $400xy(1 - x^2 - y^2)$ also has stationary points $(\pm 1, 0)$, $(0, \pm 1)$ (saddles) that the slides omit; and slide 010·4's $rt - s^2$ values are -160000 and 320000, not -1600 and 356000.
ADDED FOR CLARITY S010·3 · S010·4

Exam trap. Open box: area $xy + 2yz + 2zx$ (one base, four sides). The slide's $2xy + 2yz + zx$ and its derivative slip lead to a cube, which is wrong. ADDED FOR CLARITY S010·5 · S010·6 · T010a 00:07:38

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Free extrema of $f(x, y)$ are asked in three of them, always a 7-mark Q1 part: Dec 2023 Q1.c and Jan 2024 set 1 Q1.a ask the same function $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ (also slide 010·7 and CW problem 32), and Jan 2024 set 2 Q1.a asks $x^3 - 12x + y^3 + 3y^2 - 9y$ (CW problem 29). Both are cubic polynomials with four stationary points: one maximum, one minimum and two saddle points. The other five papers ask Lagrange's method (class 011) instead. The temperature-on-a-sphere problem of slide 010·3 is asked in June 2024 Q1.d, but by Lagrange's method; the elimination method of topic 3 gives the same answer. Trigonometric and quartic functions (topic 2) were never asked.

1 · The method, and the ESA cubic functions

ESA Dec 2023 (7 marks) UE22MA141A Q1.c · solved in W1.1

Find the extreme values of the function $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.

ESA Jan 2024 (7 marks) UE23MA141A set 1, Q1.a · solved in W1.1

Locate the stationary points of $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. Discuss their nature.

ESA Jan 2024 (7 marks) UE23MA141A set 2, Q1.a · solved in W1.2

Locate the stationary points of $f(x, y) = x^3 - 12x + y^3 + 3y^2 - 9y$. Discuss their nature.

3 · Applied problems: eliminate one variable first

ESA June 2024 (7 marks) UE23MA141A Q1.d · solved in W3.1

The temperature T at any point (x, y, z) in space is $400xyz^2$. Using Lagrange's method of multipliers, find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$. (Asked by Lagrange's method; W3.1 checks the answer by elimination.)